Variant Interpretation in CYP2C19

BIOT 6900 · Module 5 Assignment · Rakesh Vayigandla · October 2026

Gene. CYP2C19. I was not assigned a gene on the allocation list, so I chose it myself. CYP2C19 activates clopidogrel and clears proton-pump inhibitors, voriconazole and several antidepressants. **Fetch and quality checks.** The fetch returned 80 ClinVar variants (38 “other”, mostly drug response; 33 VUS; 9 B/LB; **0 P/LP**). Both anchors were present (CYP2C19*2, rs4244285; *17, rs12248560), and the score columns were populated (CADD 68/80, REVEL 54, AlphaMissense 54, gnomAD 55). **Fallbacks, disclosed.** (1) CYP2C19 has no P/LP variants, so I pooled it with DPYD to calibrate. (2) I had no AlphaGenome key, so the Atlas was skipped.

1. Thresholds and calibration

Threshold: MAX_AF = 0.5. CYP2C19 is not a disease gene. Its important alleles are common because losing CYP2C19 is harmless until a patient takes a drug that depends on it, so there is no purifying selection to keep them rare. In my data, *2 has an allele frequency of 16.8% and *17 (increased function) 20.0%. A credible maximum therefore has to sit above any functional allele, and 0.5 does so. It removed only 2 of 80 variants, rs3758581 (I331V, 95.2%) and rs17885098 (synonymous, 91.9%). Both are ClinVar Benign, and at that frequency the reference base is the minor allele. **No P/LP or drug-response variant was removed.**

The 0.001 contrast would have discarded 19 variants, including *2 (rs4244285), *17 (rs12248560), *3 (W212X, rs4986893, 0.25%), *4 (M1V, rs28399504, 0.24%), *8 (W120R, rs41291556, 0.16%) and nine more drug-response variants. In other words, a rare-disease filter would have deleted almost every allele that CPIC uses to guide clopidogrel prescribing.

Predictor: CADD. It scores every variant class (85% of my variants), whereas REVEL and AlphaMissense score only missense variants (67.5%). Most CYP2C19 function alleles are not missense (*2 is labelled synonymous, *3 stop-gained, *4 start-lost, *17 upstream), so a missense-only tool would miss the variants that matter.

Calibration was not possible on CYP2C19 alone. It has 0 P/LP and 7 B/LB variants after filtering (5 with a CADD score), so LR+ is undefined. I took the documented route and pooled with DPYD, the only one of the five genes with enough P/LP variants.

Table 1. Pooled CADD calibration (CYP2C19 + DPYD; bands of CADD phred).

Band	n	P/LP	B/LB	LR+	Evidence
0–10	65	0	65	0.00	strong, benign
10–20	24	0	24	0.00	strong, benign
20–25	11	5	6	2.05	no evidence either way
25–30	**4**	3	1	7.38	(moderate, pathogenic) **n < 10: not used**
30–50	31	31	0	∞	strong, pathogenic

What this rests on. 135 labelled variants (39 P/LP, 96 B/LB), but **130 of them are from DPYD**; CYP2C19 contributes only 5 benign variants. So this is really a DPYD calibration applied to CYP2C19. The DPYD P/LP variants all have CADD ≥ 23.4 (mean 35.5), which suggests they are mostly truncating, whereas CYP2C19’s VUS are missense. The LRs of 0 and ∞ are not exact either; they mean the band contained no variants of one label, which, with only 39 P/LP variants, could easily be chance. The 25–30 band rests on four variants, and I treated it as no evidence.

There is direct proof that the transfer is unsafe: **CYP2C19*4 (M1V, start-lost, CADD 15.9) falls in the “strong, benign” band**, yet it is a well-established no-function allele. My confidence in these bands is therefore low. They can say a variant “looks like DPYD’s benign variants”, not that it is benign in CYP2C19.

VUS classification. Of 33 VUS, **16 lean benign** (CADD ≤ 20) and **17 stay put**: 8 fall in the 20–25 band (LR 2.05, no evidence), 3 in the small 25–30 band (L87Q, G135V, R144S), and 6 are unscored structural deletions. **None move towards pathogenic.** Given the caveats above, I would describe the 16 “leaning benign” calls as weak.

2. Coverage gaps and the regulatory genome

No missense predictor could score: synonymous (7), upstream (5), intron (1), intergenic (1), splice donor (1), stop-gained (1), stop-lost (1), or the 8 large deletions, which have no consequence assigned at all. Ten variants were scored by **no** tool. A blank score here means the tool *could not see* the variant, not that it found the variant harmless.

Specific unscored variant: rs72558186 (chr10:g.94781999T>A, a splice-donor change at the intron-5 donor site; CYP2C19*7). CADD, REVEL and AlphaMissense all returned nothing, and it has no gnomAD frequency, yet ClinVar records a drug-response assertion. To classify it I would need **splicing evidence**: a SpliceAI delta score for donor loss as the in-silico line, then an **RNA assay** (a minigene, or RT-PCR from heterozygous liver or lymphoblastoid cells) to show exon skipping or intron retention and the resulting frameshift. If the assay shows a broken transcript, that is functional evidence (PS3) and would justify a no-function call far more strongly than any predictor.

The Atlas (skipped, no key). It could have scored the variants every other tool in the course misses. It would have prioritised the 25 non-missense variants. It could ask whether *17 (rs12248560, upstream) and rs7902257 (upstream, 2.2%) change **promoter activity in liver** (via DNase, ChIP and RNA-seq tracks), and whether rs4244285 and rs72558186 act through **splicing** (via MERGED_SPLICING). Nothing else in the course can do this: CADD gives non-coding variants a number but no mechanism or tissue, and REVEL and AlphaMissense cannot see them at all. I would also have to read the attribution: if conservation (CACTUS or phastCons) carried the score, the mechanism would still be unresolved.

3. A discordant variant, read mechanistically: CYP2C19*2 (rs4244285)

Rule tripped: `flag_common_but_plp`. It is common (gnomAD 16.8%) yet carries a functional ClinVar assertion (drug response).

What each tool said:

- **snpEff:** synonymous (P227P).
- **CADD:** 5.7, which falls in my “strong, benign” band.
- **REVEL and AlphaMissense:** no score.
- **ClinVar and CPIC:** the defining no-function allele, and the main reason patients get poor clopidogrel activation.

My explanation: both outputs are right about different things. For a synonymous call and a CADD of 5.7 to coexist with loss of function, the variant must act **through RNA, not protein**, and the gene must be **under weak selection**. Both are true. c.681G>A creates an **aberrant splice site in exon 5**, so the mRNA is mis-spliced, the reading frame shifts, and no functional enzyme is made. The DNA change is “silent” at the amino-acid level, which is exactly what snpEff reports and what a missense tool cannot see. CADD’s low score largely reflects that the site is a synonymous position of modest conservation. That is a fair summary for a gene where losing function costs nothing without a drug, so evolution has not removed the allele.

I prefer this explanation to the alternative, that rs4244285 is just a marker in linkage disequilibrium with a nearby causal variant. The splice defect has been shown directly, and that alternative does not explain how the allele works.

4. What I could not determine about *2

1. **Zygosity and phase.** My analysis scores single variants, but the phenotype depends on the diplotype. A *2/*2 patient is a poor metaboliser, and *2/*17 is intermediate. rs12769205 (intronic, 17.3%), which also flagged, sits on the same haplotype as *2. Without phased genotypes I cannot say which alleles are on the same chromosome.
2. **Modifiers, including phenoconversion.** A patient’s real CYP2C19 activity also depends on co-medication. Omeprazole and fluvoxamine inhibit CYP2C19 and can make a genotypic normal metaboliser behave like a *2 carrier. Nothing in a variant table can capture this.

The single experiment. Allele-specific RT-PCR (or long-read RNA sequencing) of CYP2C19 transcripts from **human liver samples heterozygous for rs4244285**. If the A allele produces the aberrant exon-5 transcript and the G allele does not, the splicing mechanism is confirmed. **What would change my mind:** if both alleles spliced normally, the effect must come from a linked variant (rs12769205 is the obvious candidate), and I would re-assign the causal call to it.

References

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